

STATISTICS

PART 6: Sampling Theory (for Two Populations)

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Sampling Theory

Statistics

2

Sampling Distribution for Two Populations

The comparison of parameters between two population constitutes an important class of problems in statistics. In this context, the following parameters will be analyzed:

- the difference between population means $\rightarrow \mu_1 - \mu_2$
- the difference between population proportions $\rightarrow p_1 - p_2$
- The ratio of population variances $\rightarrow \frac{\sigma_1^2}{\sigma_2^2}$

Sampling Theory

Statistics

3

Sampling Theory for the Difference between Means

Consider a sample of size n_1 whose members are

$$X_{11}, X_{12}, \dots, X_{1n_1}$$

is selected from a population with mean μ_1 and variance σ_1^2 , and similarly another sample of size n_2 whose members are

$$X_{21}, X_{22}, \dots, X_{2n_2}$$

is selected from a population with mean μ_2 and variance σ_2^2 . Then,

$$E(\bar{X}_1 - \bar{X}_2) = E(\bar{X}_1) - E(\bar{X}_2) = \mu_1 - \mu_2,$$

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = (1)^2 \text{Var}(\bar{X}_1) + (-1)^2 \text{Var}(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Sampling Theory

Statistics

Theorem: If the two populations are distributed normally with the means μ_1 and μ_2 , and the variances σ_1^2 and σ_2^2 , then the random variable

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a standard normal distribution, where n_1 and n_2 are sample sizes.

Proof: Since $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$,

$$\bar{X}_1 \sim N\left(\mu_1, \sigma_1^2/n_1\right) \text{ and } \bar{X}_2 \sim N\left(\mu_2, \sigma_2^2/n_2\right),$$

then we have

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \sigma_1^2/n_1 + \sigma_2^2/n_2\right)$$

$$\Rightarrow Z = \frac{(\bar{X}_1 - \bar{X}_2) - E(\bar{X}_1 - \bar{X}_2)}{\sqrt{\sigma_{\bar{X}_1 - \bar{X}_2}^2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Sampling Theory

Statistics

5

Sampling Theory for the Difference between Means (for unknown variances)

Theorem: If the two populations are distributed **normally** with the means μ_1 and μ_2 , and their variances are equal, i.e., $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (but their values are unknown), then the random variable

$$t_{n_1+n_2-2} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t distribution with degree of freedom $n_1 + n_2 - 2$. Here,

$$S_k = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

Sampling Theory

Statistics

6

Proof: Since $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, the random variables

$$\chi_1^2 = \frac{(n_1 - 1)S_1^2}{\sigma_1^2} \text{ and } \chi_2^2 = \frac{(n_2 - 1)S_2^2}{\sigma_2^2}$$

have Chi-Square distribution with the degrees of freedom $n_1 - 1$ and $n_2 - 1$. If we take $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then the random variable

$$Y = \chi_1^2 + \chi_2^2 = \frac{(n_1 - 1)S_1^2}{\sigma^2} + \frac{(n_2 - 1)S_2^2}{\sigma^2} = \frac{(n_1 + n_2 - 2)S_k^2}{\sigma^2}$$

has also Chi-Square distribution with the degrees of freedom $n_1 + n_2 - 2$ where

$$S_k = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

Finally, from $t_v \sim \frac{Z}{\sqrt{\chi_v^2/v}}$,

$$\frac{Z}{\sqrt{\chi_{n_1+n_2-2}^2/(n_1+n_2-2)}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} \xrightarrow{\sim t_{n_1+n_2-2}} \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Sampling Theory

Statistics

7

Theorem: If the two populations are distributed **normally** with the means μ_1 and μ_2 , and their variances are not equal, i.e., $\sigma_1^2 \neq \sigma_2^2$ (but their values are unknown), then the random variable

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

has a t distribution with degree of freedom ν which can be computed by rounding of

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Sampling Theory

Statistics

8

Sampling Theory for the Difference between Means (for unknown populations)

According to the **Central Limit Theorem**; if two populations has any distribution with the means μ_1 and μ_2 , and the variances σ_1^2 and σ_2^2 , then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$. Also, if the population variances σ_1^2 and σ_2^2 are unknown, then the random variable

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

has approximately the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$.

Sampling Theory

Statistics

9

Sampling Theory for the Difference between Means (For Paired Populations)

Up to this point, we have investigated the difference between the means of two **independent** populations. However, when the measurements are paired (or repeated), the populations become **dependent**.

For example:

- The weight of people before and after a diet program
- The level of a blood parameter before and after taking a new drug
- The prices of a group of some specific products sold in two different shopping centers

In cases of dependent populations, such as the examples above, we should use a different technique to estimate the difference between population means.

Sampling Theory

Statistics

10

For paired populations:

Let n paired members be chosen from two dependent populations with random variables X_1, X_2 and means μ_1, μ_2 . Then,

$$D_i = X_{1i} - X_{2i}, \quad i = 1, 2, 3, \dots, N$$

is a new random variable that represents of difference between paired members, where N is the size of both populations . Thus, the population mean will be

$$\mu_D = \frac{\sum_{i=1}^N D_i}{N}$$

and the sample mean and sample standard deviation will be

$$\bar{D} = \frac{\sum_{i=1}^n D_i}{n} \quad S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n - 1}}$$

Moreover

$$E(\bar{D}) = E\left(\frac{\sum_{i=1}^n D_i}{n}\right) = \frac{1}{n} \sum_{i=1}^n E(D_i) = \frac{1}{n} n \mu_D = \mu_D$$

Sampling Theory

Statistics

11

Theorem: Let a sample of size n is chosen from two **paired** populations which are distributed **normally** and μ_D is the population mean of $D_i = X_{1i} - X_{2i}$. If $\bar{D}$ and S_D are the sample mean and sample standard deviation, respectively, then the random variable

$$t_{n-1} = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}}$$

has a t distribution with the degree of freedom $n - 1$.

According to the **Central Limit Theorem**; for paired populations having unknown distribution, $\bar{D}$ has normal distribution and the random variable

$$z = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}}$$

has the standard normal distribution.

Sampling Theory

Statistics

12

Sampling Theory for the Difference between Proportions

Let two samples with size n_1 and n_2 are from two populations, respectively. Let X_1 and X_2 be random variables related to these two populations.

If $\hat{p}_1 = \frac{X_1}{n_1}$ and $\hat{p}_2 = \frac{X_2}{n_2}$, then

$$E[\hat{p}_1 - \hat{p}_2] = E[\hat{p}_1] - E[\hat{p}_2] = p_1 - p_2$$

and

$$\text{Var}[\hat{p}_1 - \hat{p}_2] = (1)^2 \text{Var}(\hat{p}_1) + (-1)^2 \text{Var}(\hat{p}_2) = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}.$$

Sampling Theory

Statistics

13

According to the **Central Limit Theorem**; $(\hat{p}_1 - \hat{p}_2)$ has normal distribution and the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - E[\hat{p}_1 - \hat{p}_2]}{\sqrt{\text{Var}[\hat{p}_1 - \hat{p}_2]}} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

has the standard normal distribution for $n_1 \geq 30$ and $n_2 \geq 30$.

In practice, for the large values of n_1 and n_2 , we can use $\hat{p}_1$ and $\hat{p}_2$ instead of p_1 and p_2 . Thus, the random variable

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}}$$

has also approximately distributed standard normally.

Sampling Theory

Statistics

14

Sampling Theory for the Ratio of Two Variances

Theorem: If S_1^2 and S_2^2 are the variances of the independent random samples of size n_1 and n_2 which are selected from two populations distributed normally with variances σ_1^2 and σ_2^2 , the random variable

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2}$$

has a F distribution with the degree of freedoms $n_1 - 1$ and $n_2 - 1$.

Sampling Theory

Statistics

Proof:

We know that $F_{v_1, v_2} \sim \frac{U/v_1}{V/v_2}$ if $U \sim \chi_{v_1}^2$ and $V \sim \chi_{v_2}^2$ are independent variables.

Also, since populations have normal distributions, we can write:

$$\chi_{n_1-1}^2 \sim \frac{(n_1-1)S_1^2}{\sigma_1^2} \text{ and } \chi_{n_2-1}^2 \sim \frac{(n_2-1)S_2^2}{\sigma_2^2}.$$

$$\Rightarrow F_{n_1-1, n_2-1} \sim \frac{\chi_{n_1-1}^2 / (n_1 - 1)}{\chi_{n_2-1}^2 / (n_2 - 1)} = \frac{\frac{(n_1 - 1)S_1^2}{\sigma_1^2(n_1 - 1)}}{\frac{(n_2 - 1)S_2^2}{\sigma_2^2(n_2 - 1)}} = \frac{\frac{(n_1 - 1)S_1^2}{\sigma_1^2(n_1 - 1)}}{\frac{(n_2 - 1)S_2^2}{\sigma_2^2(n_2 - 1)}} = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

Sampling Theory

Statistics

16

OUTLINE: Theorems for Two Independent Populations

